With the above adjustments, our semi-discrete high-order Lagrangian type DG scheme can be written as follows: find $\mathbf{U} \in \mathbf{V}_h^k(t)$ to satisfy the following semi-discrete scheme

$$\frac{d}{dt}\iint_{I_{ij}(t)} \mathbf{U} dxdy = -\int_{\partial I_{ij}(t)\setminus\Gamma_{cs}^{ij}} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}}-\mathbf{u}_g)\cdot\mathbf{n}\right) ds - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}}\cdot\mathbf{n}\right) ds \tag{2.9}$$

$$\begin{aligned}\frac{d}{dt}\iint_{I_{ij}(t)} (\mathbf{U}\phi_m)\, dxdy &= \iint_{I_{ij}(t)} \left(\mathbf{U}\left(\mathbf{u}\cdot\nabla\phi_m + \frac{\partial\phi_m}{\partial t}\right)\right) dxdy \\ &\quad - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}}-\mathbf{u}_g)\cdot\mathbf{n}\phi_m\right) ds \\ &\quad + \iint_{I_{ij}(t)} (\mathbf{G}\cdot\nabla\phi_m) dxdy - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}}\cdot\mathbf{n}\phi_m\right) ds,\end{aligned} \tag{2.10}$$

where $m = 1, \cdots, K$ and $K = \frac{(k+1)(k+2)}{2} - 1$. Let $\phi_0 = 1$, $\{\phi_0, \phi_1, \cdots, \phi_K\}$ are the $K+1$ basis functions in the DG space V_h^k, and $\{\phi_1, \cdots, \phi_K\}$ are orthogonal to $\phi_0 = 1$. We take $\Gamma_{cs}^{ij} = \partial I_{ij} \cap \{\Gamma_c \cup \Gamma_s\}$. Γ_c is the set of cell edges containing the contact discontinuities and material interfaces, and Γ_s is the set of cell edges containing shock waves.

Cell_moments_array $Q[m, j, k] = \int \phi_m U_j$ in cell k

(ϕ_m, U_j, cell k)

fix basis

$$\text{Volume_rhs} := \iint \left[\bar{U}\left(u\cdot\nabla\phi_m + \frac{\partial\phi_m}{\partial t}\right) + G\cdot\nabla\phi_m\right] dxdy$$

in 2D

$$\iint \begin{bmatrix} U \\ \vdots \end{bmatrix} \left(u_x\cdot\partial_x\phi_m + u_y\cdot\partial_y\phi_m + \partial_t\phi_m\right) + \begin{pmatrix} 0 & 0 \\ P & 0 \\ 0 & P \\ Pu_x & Pu_y \end{pmatrix} \begin{pmatrix} \phi_x \\ \phi_y \end{pmatrix} dxdy$$

---

fix i, k $\quad Q(:, i, k)$ we need (source) | Cell motion | geometry | options

$$= \sum_{i=1}^{n(q_i)} \left[u_i \begin{pmatrix} \left(u_x\partial_x\phi_1(q_i) + u_y\partial_y\phi_{nb}(q_i) + \partial_t\phi_1(q_i) + G_1[i,1]\partial_x\phi(q_i) + G_1[i,2]\partial_y\phi(q_i)\right) \\ \vdots \\ \left(u_x\partial_x\phi_{nb}(q_i) + u_y\partial_y\phi_{nb}(q_i) + \partial_t\phi_m(q_i) + G_1[i,1]\partial_x\phi(q_i) + G_1[i,2]\partial_y\phi(q_i)\right) \end{pmatrix} W(q_i) \right]$$

for all q_i

Use coefficient to get geo.eval_cons $\begin{bmatrix} U_1 \\ \vdots \\ U_4 \end{bmatrix}(q_i)$, $G(U) = [\ \ |\ \ ]$

basis_at_quadrature_mat(:, pt_index) — d_x, d_y

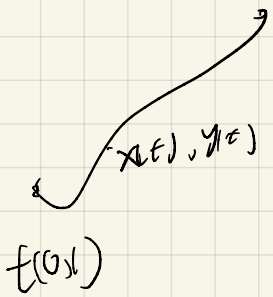
flux_x flux_y

$$\left[U_x \begin{bmatrix} \partial_x \phi_1(q_1) \\ \vdots \\ \partial_x \phi_{nb}(q_1) \end{bmatrix} + U_y \begin{bmatrix} \\ \\ \\ \end{bmatrix} + \begin{bmatrix} \partial_t \phi \end{bmatrix} \right]$$

for $U_1 \cdots U_4$

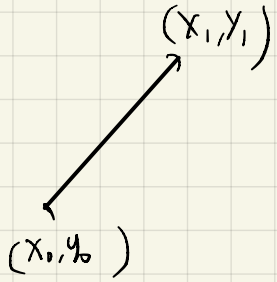
$$W_i * U_i \left(\left[\sim \right] + flux_x(i) \cdot \left[\partial_x \phi \right] + flux_y(i) \left[\partial_y \phi \right] \right)$$

updated ✓



$$\int_0^1 f(x(t), y(t))\, ds$$

straight



$$\int_0^1 f(x(t), y(t)) \sqrt{x'(t)^2 + y'(t)^2}\, dt$$

$$\Downarrow$$

$x(t) = x_0 + (x_1 - x_0)t$

$y(t) = y_0 + (y_1 - y_0)t$

$x'(t) = (x_1 - x_0)$

$y'(t) = (y_1 - y_0)$

$$\Rightarrow \sqrt{x'(t)^2 + y'(t)^2} = |\text{length}| \cdot \int_0^1 f(x(t), y(t))\, dt$$

$$= |\text{length}| \cdot \sum f(q_i) w_i$$

Given All date in some time

We want to compute flux in (2,12) (2,13)

$G(\hat{U})$ **Face flux**

$$Flux = \begin{bmatrix} 0 & 0 \\ P_* & 0 \\ 0 & P_* \\ P_* v_{x*} & P v_{y*} \end{bmatrix} + \hat{U}(v_* - u_g \cdot n)$$

$$\begin{cases} \hat{G} \cdot n = (0, P_* n, P_* u_*)^T \\ \hat{U} = (\hat{P}, \hat{Pu}, \hat{E})^T = \begin{cases} P^{in}, P^{in} u^{in}, e^{in} & u_* - u_g \cdot n \geq 0 \\ P^{out}, P^{out} u^{out}, e^{out} & u_* - u_g \cdot n < 0 \end{cases} \end{cases} \Rightarrow Flux_2 = \hat{U} \cdot (v_* - u_g \cdot n$$

we neeed to build a fanction: Given a face, we want to get flux at (x,y)

So we need get P_*, u_*. To do that, we need U_L, U_R

① Interior pts: face_cons

first we need compute U_L, U_R (r_L r_R) at given point_coord (x,y)

face — left cell index, right cell index

$U_L(x) =$ Geo. eval_cons

Similary for period.

Cells_coefficients
Cells_geometry
point_coord,
expansion_fs

$U_R(x)$ - -

② boundary:

So we define Outside_cons

to generate Fake outside value c

if : wall fake = Euler.reflect(inside, normal)

if : fixed ~ initial_cons(problem, coords)

if overflow copy ~.

Given face ... Coord get U_L, U_R

U_L is easy since all of our face has read left

Using geo.eval( )

if U_R is boundary < periodic same / others

After we have U_L, U_R Give $\vec{v_g}$

HLLC → W^*, p^*

$G(\hat{U})$

$$Flux = \begin{bmatrix} 0 & 0 \\ P_* & 0 \\ 0 & P_* \\ P_* v_{x*} & P_* v_{y*} \end{bmatrix} + \hat{U}(v_* - v_g \cdot n)$$

return pressure flux

$$\hat{G} \cdot n = (0, P_* n, P_* u_*)^T$$

$$\hat{U} = (\hat{\rho}, \hat{\rho u}, \hat{E})^T = \begin{cases} \rho^{in}, \rho^{in} u^{in}, e^{in} & u_* - v_g \cdot n \geq 0 \\ \rho^{out}, \rho^{out} u^{out}, e^{out} & u_* - v_g \cdot n < 0 \end{cases}$$

$$\Rightarrow Flux = \hat{U} \cdot (v_* - v_g \cdot n)$$

return advection flux

return

$P_jump = P^{-up} - P^{+down}$

Flux

With the above adjustments, our semi-discrete high-order Lagrangian type DG scheme can be written as follows: find $\mathbf{U} \in \mathbf{V}_h^k(t)$ to satisfy the following semi-discrete scheme

$$\frac{d}{dt}\iint_{I_{ij}(t)} \mathbf{U} dxdy = -\int_{\partial I_{ij}(t)\setminus\Gamma_{cs}^{ij}} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}}-\mathbf{u}_g)\cdot\mathbf{n}\right) ds - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}}\cdot\mathbf{n}\right) ds \tag{2.9}$$

$$\begin{aligned}\frac{d}{dt}\iint_{I_{ij}(t)} (\mathbf{U}\phi_m)\, dxdy = &\iint_{I_{ij}(t)} \left(\mathbf{U}\left(\mathbf{u}\cdot\nabla\phi_m + \frac{\partial\phi_m}{\partial t}\right)\right) dxdy \\ &- \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}}-\mathbf{u}_g)\cdot\mathbf{n}\phi_m\right) ds \\ &+ \iint_{I_{ij}(t)} (\mathbf{G}\cdot\nabla\phi_m)\, dxdy - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}}\cdot\mathbf{n}\phi_m\right) ds,\end{aligned} \tag{2.10}$$

where $m = 1, \cdots, K$ and $K = \frac{(k+1)(k+2)}{2} - 1$. Let $\phi_0 = 1$, $\{\phi_0, \phi_1, \cdots, \phi_K\}$ are the $K+1$ basis functions in the DG space V_h^k, and $\{\phi_1, \cdots, \phi_K\}$ are orthogonal to $\phi_0 = 1$. We take $\Gamma_{cs}^{ij} = \partial I_{ij} \cap \{\Gamma_c \cup \Gamma_s\}$. Γ_c is the set of cell edges containing the contact discontinuities and material interfaces, and Γ_s is the set of cell edges containing shock waves.

for each edge I calculate flux and save it first.

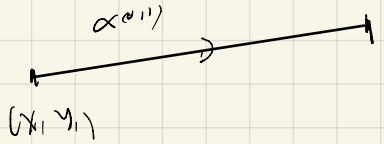
$$\int_{\partial I_{ij}} \begin{pmatrix}\hat{U}_1 \\ \vdots \\ \hat{U}_4\end{pmatrix}(\hat{u}-u_g)\cdot n\,\phi_m = \int_{\partial I_{ij}} \text{advection flux}\cdot\phi_m$$

$$\int_{\partial I_{ij}} (\hat{G}\cdot n)\phi_m ds = \int_{\partial} \text{pressure flux}\ \phi_m$$

eval_face_flux → return Tuple

edge

quadrature $(\alpha, 1)$



$$\alpha\left[(x_2,y_2)-(x_1,y_1)\right]+(x_1,y_1)$$

$$(1-\alpha)(\bar{x}_1^2)+\alpha\,\bar{x}_2^2$$

find $\hat{U}$ at each $q_i = \alpha \vec{x}_{start} + (1-\alpha)\vec{x}_{end}$

edges containing the shocks. In this paper, we apply the KXRCF detection technique [23] to identify these edges. Let l be a cell edge. Define u_h^- and u_h^+ as the upwind ($(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n} > 0$) and downwind ($(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n} < 0$) sides of the indicator variable u_h along l, respectively. The cell edge l is marked in the set Γ_s when shock detector

$$\frac{|\int_l (u_h^- - u_h^+) ds|}{|l|^{\frac{k+3}{2}} \|u_h^-\|_{L^\infty(l)}} > C_k, \tag{2.8}$$

where C_k is a constant, usually, we take $C_k = 1$. $\|\cdot\|_{L^\infty(l)}$ is the standard L^∞ norm on the edge l. Here we take the pressure p as the indicator variable. When $l \in \Gamma_s$, we ignore this flux term $\int_l (\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n})\, ds$ in the cell average equations, similar to the treatment at the material interfaces.

detection $\dfrac{|\text{length}| \sum_{q_i} (p^- - p^+)(q_i)\, w_i}{|l|^{\frac{k+3}{2}} \max_{q_i} |p^-(q_i)|} > 1$ (p jump) shock

it's so... sloppy

---

for all faces

flux$_1$ ... flux$_4$ $\begin{pmatrix} \phi_1 \\ \vdots \\ \phi_m \end{pmatrix}$ if keep-mean

for in in q_i

left_cell $+=$ $\sum_{q_i}$ advection flux$(q_i)\, \phi_m(q_i)\, w_i$

right_cell $+=$ $\sum_{q_i}$ pressure flux $\phi_m(q_i)\, w_i$